IV. The dirt is lifted off the surface of the material upon shaken and suspended in water.

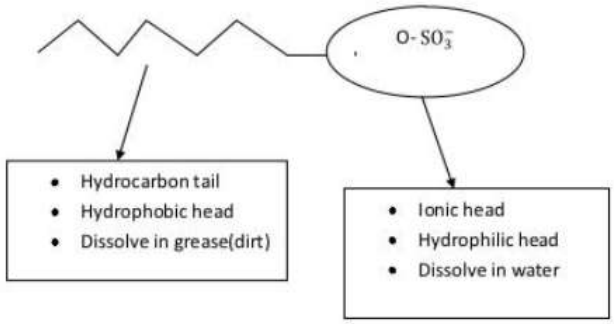
V. The tail region emulsifies and breaks up the grease into small drops.

VI. When shaken, the water molecules will attract the soap ions and cause the dirt to detach from the surface of the material.

VII. The soap bubbles help to float the dirt emulsion in the water. When rinsed, the dirt will be removed together with water.



- The cleansing action of detergent and soap is fundamentally same.
- When a detergent dissolves in water, its molecule will dissociate to form sodium or potassium ion and detergent ion (detergent anion).
- The detergent ion such as R-O- SO_3^- can be represented in a simplified form by structure below.



Cleansing action of detergents

I. Detergent molecule dissociate to form sodium or potassium ions and detergent ions.

II. Hydrophilic region dissolve in water. Hydrophobic dissolve in grease and emulsifies the grease.

III. When the water is shaken, the detergent ion will remove the dirt.

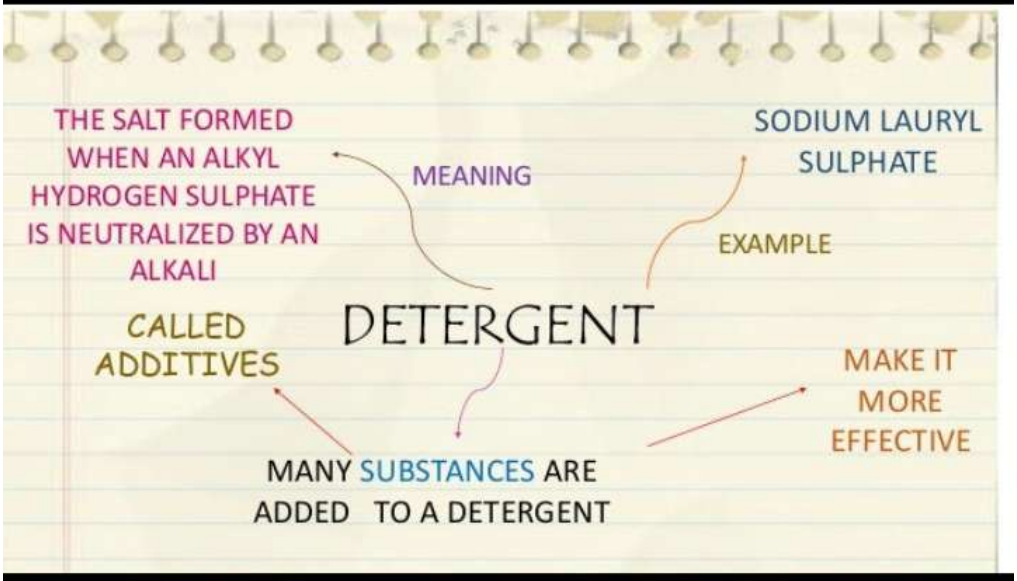
g) Compare and contrast the effectiveness of cleansing action of soap and detergent in hard water by using mid map or table.

- **Hard water** contains a great amount of calcium and magnesium ions.
- **Soap** cannot be used in hard water because the magnesium and calcium ions wil react with soap ions to form a non-soluble precipitate in water that is called a soap film or scum. Soap cannot produce bubble in hard water.
- **Soap scum** is not easily cleaned. It will leave a clear sediment that can be seen on the clothes and cause the clothes to feel hard.

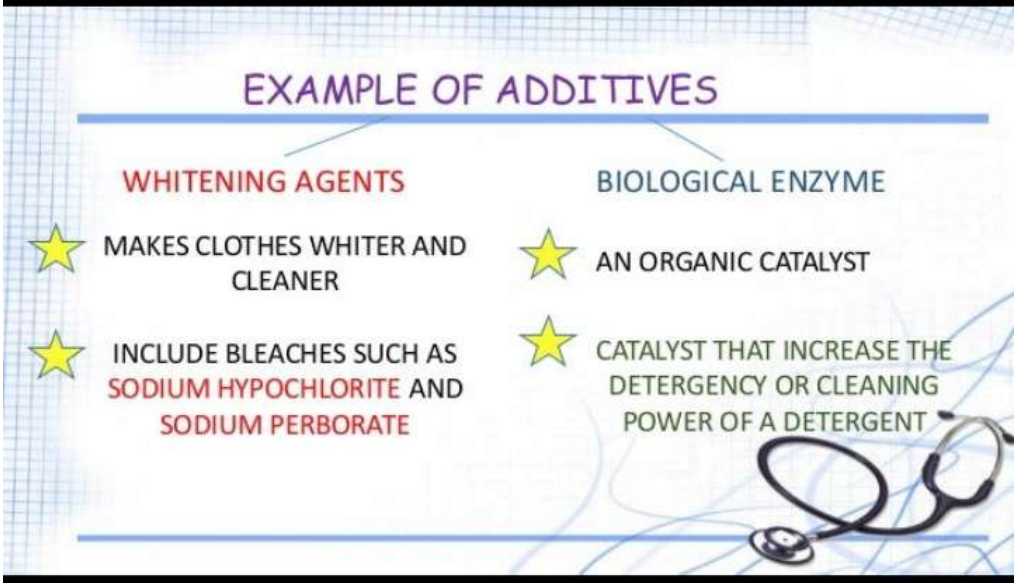
Properties	Soap	Detergent
Cleaning power	Less powerfull	More powerfull
Ease of rinsing	It is difficult to wash away all soap on clothes. The soap that remains leave and odour and spoils the fabrics	Rinse out well from clothes
pH	Slightly alkaline	Can be controlled to suit the cleaning task
Molecular structure	Determined by the fatty acids found in the oil or fat used to produce soap	Can be modified to suit the cleaning task. For example, a detergent can be made specially for cleaning toilet bowls
Formation of scum	Form scum with hard water	Does not form scum with hard water

h) Detergent consists of various addictives. What is the purpose of adding detergent addictives ? Construct a mind map to show the addictives in detergents and its functions.

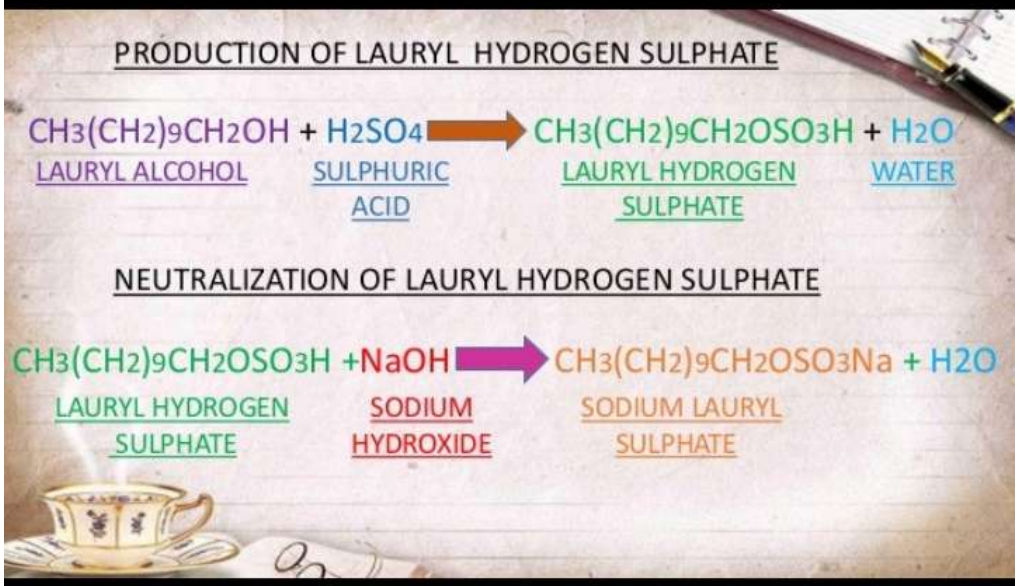
Addictives	Example	Function
Biological enzyme	Amylases, proteases, cellulases	To remove protein stains such as blood
Whitening agent	Sodium perborate	To convert stains into colourless substances
Optical whitener	Fluorescent dyes	To add brightness and whiteness to white fabrics
Builder	Sodium tripolyphosphate	To enhance the cleaning efficiency of detergent by softening the water
Suspension agent	carboxymethylcellulose	To prevent the dirt particles removed from redepositing onto cleaned fabrics.
Filler	Sodium sulphate, sodium silicate	To add to tha bulk of the detergents and enable it to be poured easily
Foam control agents	Silicones	To control foaming in detergent
Fragrance	-	To add fragrance to both detergent and fabrics



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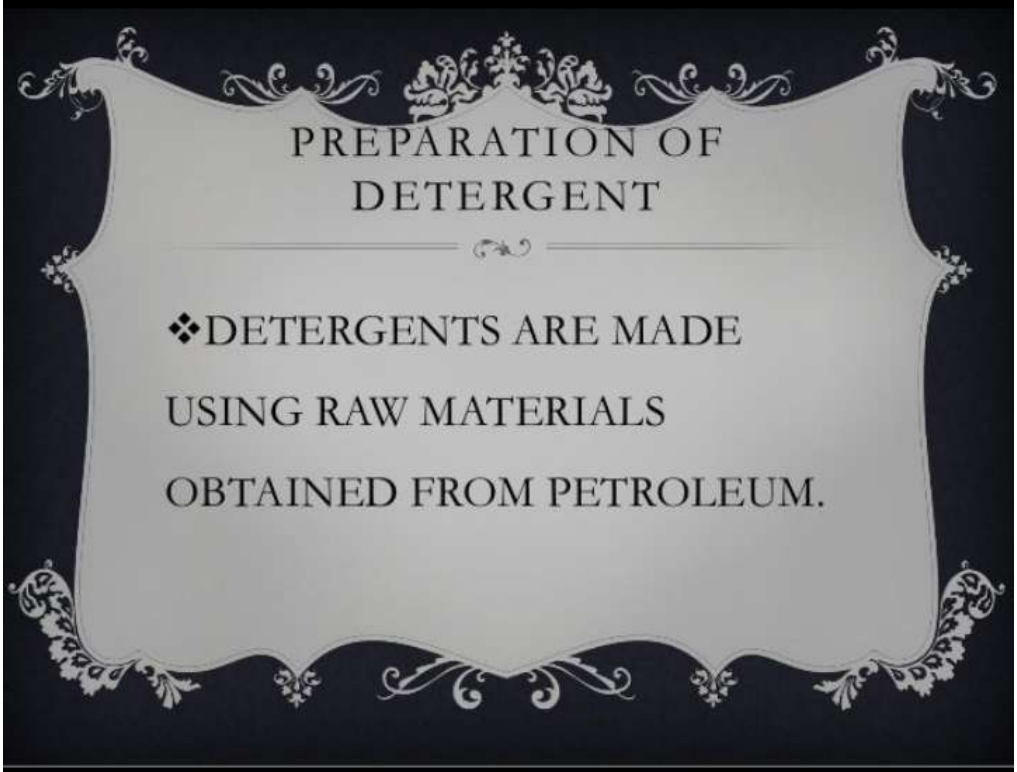
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PREPARATION OF
DETERGENT
❖DETERGENTS ARE MADE
USING RAW MATERIALS
OBTAINED FROM PETROLEUM.

Differences between soaps and detergents

- **SOAPS**
  - They are metal salts of long chain higher fatty acids.
  - These are prepared from vegetable oils and animal fats.
  - They cannot be used effectively in hard water as they produce scum i.e., insoluble precipitates of Ca^{2+}, Mg^{2+}, Fe^{2+} etc.
- **DETERGENTS**
  - These are sodium salts of long chain hydrocarbons like alkyl sulphates or alkyl benzene sulphonates.
  - They are prepared from hydrocarbons of petroleum or coal.
  - These do not produce insoluble precipitates in hard water. They are effective in soft, hard or salt water.

Soap	Detergent
• Its effectiveness is reduced when used in hard water.	• Is effective even in hard water.
• Cannot be used in hard water because the magnesium and calcium ions will react with the soap anions to form a non-soluble precipitate in water called scum.	• Does not form scum in hard water because the ionic head reagion of the detergent will not form a strong ionic bond with the magnesium and calcium ions.
$RCOONa(s) \xrightarrow{H_2O} RCOO^-(aq) + Na^+(aq)$ Soap Solid Soap $RCOO^-(aq) + Ca^{2+}(aq) \rightarrow (RCOO)_2Ca(s)$ Soap Calcium Ion Precipitate(scum) $RCOO^-(aq) + Mg^{2+}(aq) \rightarrow (RCOO)_2Mg(s)$ Soap Magnesium Ions Precipitate(scum)	$[R\text{-}O\text{-}SO_3^-]_2Ca^{2+} \rightarrow 2R\text{-}O\text{-}SO_3^- + Ca^{2+}$ $[R\text{-}O\text{-}SO_3^-]_2Mg^{2+} \rightarrow 2R\text{-}O\text{-}SO^{3-} + Mg^{2+}$
• Soap cannot produce bubbles in hard water and this condition results in the wastage of soap.	• Detergent forms bubbles in both soft and hard water. This makes The detergent more effective in its cleansing action compared to soap.
• Scum is not easily cleaned. It will leave a clear sediment that can be seen on clothes , and causes them to feel hard. • In addition, scum will stick to the inner parts of the sink and washing machine	• Calcium or magnesium salts produced, easily dossociate in water.

WHAT ARE DETERGENTS?

A **detergent** is a surfactant or a mixture of surfactants with **cleaning properties in dilute solutions**. These substances are usually alkylbenzenesulfonates, a family of compounds that are similar to soap but are more soluble in hard water.

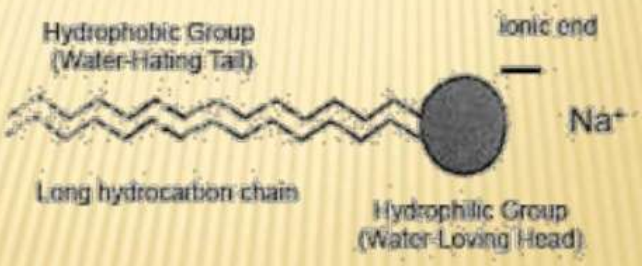


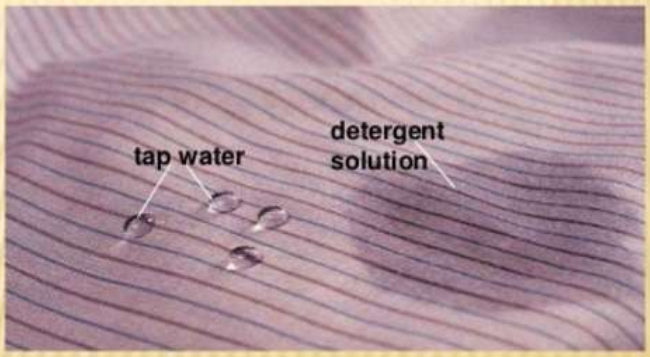
Fig 3. A detergent or soap molecule

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PROPERTIES OF DETERGENTS

1. As a wetting agent

A detergent reduces the surface tension of water. As a result of this, water spreads over the surface and wets it more easily. A detergent thus acts as a wetting agent.

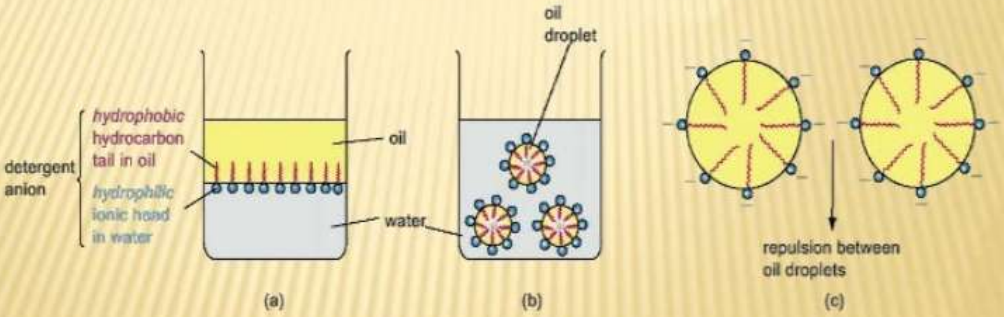


A detergent increases the wetting power of water. Tap water does not wet this piece of cloth easily, but a detergent solution does.

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2. As an emulsifying agent

Oil and water do not mix. An oil-water emulsion is unstable. On standing, the tiny oil droplets rapidly join together and grow larger to form a separate oily layer again.

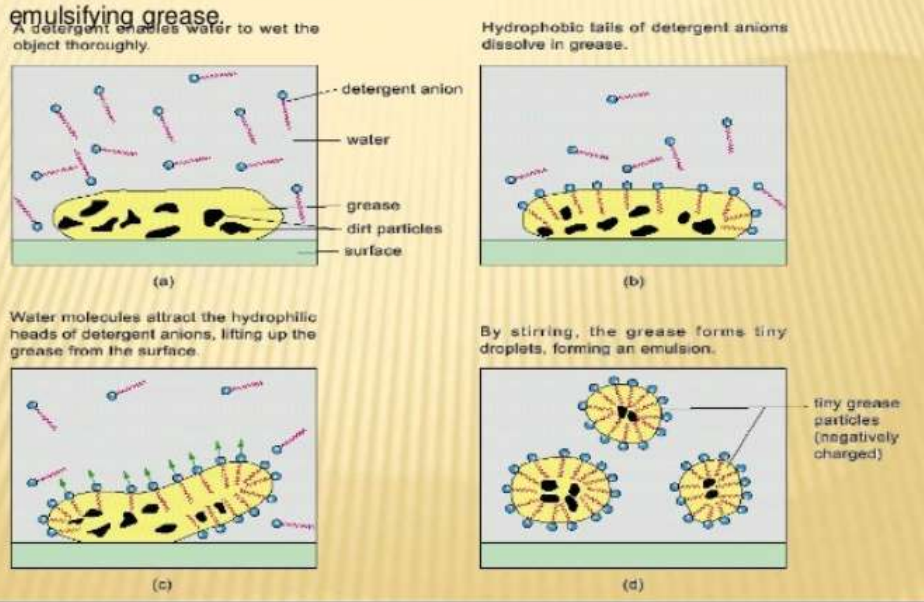


An oil-water emulsion is stabilized by a detergent.

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CLEANSING ACTION OF DETERGENTS

DETERGENTS are cleansing agents. They work by reducing the surface tension of water, enabling it to wet things more effectively, and by emulsifying grease.

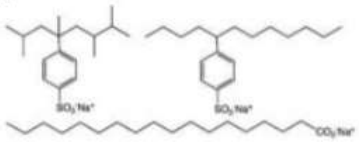


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Chemistry of detergent taskeen rubab
TaskeenRubab

Detergents

- Detergents are the sodium salt of long chain benzene sulphuric acids
- Detergents are primarily surfactants, which could be produced easily from petrochemicals
- Surfactants lower the surface tension of water, essentially making it wetter so that that it is less likely to stick to itself and more likely to interact with oil and grease
- The ionic group is in detergent is SO2Na



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Chemistry of detergent taskeen rubab
TaskeenRubab

Introduction

A detergent is a surfactant or mixture of surfactants with purifying properties in dilute solutions. In the domestic context, the term detergent itself refers specifically to laundry detergent or dish detergent, as opposed to hand soap or other types of cleaning agents.



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How do detergents work

Soaps and detergents are made from long molecules that have a head and tail. These molecules are called tensioactives; a surfactant molecule is described in the diagram below. The head of the molecule is drawn to water (hydrophilic) and grease and dirt attract the tail (hydrophobic).



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Advantages of Detergents

- Since detergents are the salts of strong acids they do not decompose in acidic medium
- Thus detergents can effectively clean fabric even if the water is acidic
- Synthetic detergents are more soluble in water than soaps
- They have a stronger cleansing action than soaps
- As detergents are derived from petroleum they save on natural vegetable oils, which are important as essential cooking medium

Disadvantages of detergents

- Many detergents are resistant to action of the action of biological agents and thus are not biodegradable. Their elimination from municipal wastewaters by the usual treatments is a problem
- They have a tendency to produce stable foams in rivers that extend over several hundred meters of the river water.
- This is due to effects of surfactants used in their preparation. Thus they pose a danger to aquatic life
- They tend to inhibit oxidation of organic substances present in wastewaters because they form a sort of envelop around them